

SRI KRISHNA KALYAN BANDARU

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ABOUT ME

AI and software graduate with a Master's in Artificial Intelligence & Machine Learning (University of Adelaide) and an engineering background spanning Mechatronics, control systems, and mechanical design. I build things end-to-end — from writing and testing code to deploying on cloud infrastructure, edge hardware, and autonomous systems. I've worked on IoT-adjacent drone platforms, embedded AI on microcontrollers, and production ML pipelines. Comfortable across Python, Docker, CI/CD, ROS2, and cloud environments, and I enjoy making complex systems actually work outside the lab. Looking to apply that in a technical role where software, data, and intelligent systems intersect — and where shipping things that actually work is what counts.

EDUCATION

Master of Artificial Intelligence & Machine Learning Dec 2025

University of Adelaide | **GPA: 6.1** | Focus: Computer Vision, Deep Learning, NLP, MLOps

Master of Science in Mechatronics Engineering 2022

University of Rome 'Tor Vergata' | **Grade: 106/110** | Focus: Control Systems, MATLAB/Simulink

Bachelor of Mechanical Engineering 2018

KL University, India | **Grade: First Class with Distinction** | Focus: Mechanical System Design, Engineering Fundamentals, C Programming, Java, Data Structures

TECHNICAL SKILLS

Languages	Python, C/C++, Java, SQL, MATLAB/Simulink, Git
AI & ML	YOLO-Seg, Mask-RCNN, BERT, Transfer Learning, TensorFlow, PyTorch, Scikit-learn
Deployment & Cloud	Docker, FastAPI, CI/CD Pipelines, Azure, AWS, Linux, Jetson Nano, STM32
Systems & IoT	ROS2, Gazebo, ArduPilot SITL, Embedded Systems, Edge AI, Autonomous Navigation
Data & Tools	pandas, numpy, Tableau, Roboflow, Streamlit, Agile/DevOps, SAP PM

EXPERIENCE

Research Assistant — Auto-ID Lab, University of Adelaide Feb 2025 – Aug 2025

Adelaide, Australia

- Migrated autonomous UAV codebase from ROS1 to ROS2, managing version control, integration testing, and system compatibility across the full stack
- Containerised and deployed AI workloads onto Jetson Nano edge hardware — managed full infrastructure integration from dev environment to live drone systems
- Built and maintained Gazebo simulation environment for testing and validating autonomous obstacle avoidance before real-world deployment
- Developed automated control algorithms with monitoring capabilities for drone flight operations, contributing to CI/CD pipelines for iterative testing and deployment

Machine Learning Engineer Intern — Proto Corp Pvt Ltd Aug 2024 – Nov 2024

Remote, India

- Deployed production ML segmentation models (YOLO-Seg, Mask-RCNN) to live servers, optimising inference performance and reliability under real load
- Built a scalable FastAPI backend for automated model serving, reducing manual effort across the deployment lifecycle
- Designed end-to-end data pipelines for collection, labelling, and model retraining using Roboflow
- Built a Streamlit testing interface that made internal model evaluation faster and more accessible for the team

Project Engineer — Wipro Technologies May 2018 – Sept 2019

India

- Managed plant operations using SAP PM, implementing process automation to reduce downtime and improve operational efficiency
- Led a team of four across cross-functional technical teams, delivering to quality and timeline standards

KEY PROJECTS

Autonomous UAV Object Detection System for Disaster Response

- Built an autonomous drone navigation system combining Ego Planner for obstacle avoidance with YOLO-based real-time object detection — full end-to-end from design to deployment
- Integrated ArduPilot SITL flight controller with Gazebo for hardware-in-the-loop testing and validation
- Developed modular ROS2 architecture connecting perception, planning, and control modules across the system

Embedded Neural Network on STM32 Microcontroller (On-Device AI)

- Deployed a computer vision model directly onto a microcontroller achieving **97% accuracy** across 18 categories — no cloud, fully on-device inference
- Applied model compression and weight quantisation using STM32CubeMX and X-CUBE-AI to fit within hardware memory constraints

Deep Learning for Alzheimer's Disease Diagnosis from MEG Signals

- Built a deep learning pipeline using transfer learning (AlexNet, ResNet, GoogLeNet) achieving **81% accuracy** in predicting Alzheimer's progression from neurological data
- Engineered feature extraction workflow converting MEG time-series into 2D representations for SVM and LDA classification, with AUC of 0.74 for early-stage diagnosis